Anchit Sharma

Data Scientist | Credit Risk |Python | SQL | Machine Learning

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Summary:

- Developed regulatory-compliant credit risk models under IND AS 109 (IFRS 9) for Expected Credit Loss (ECL) computation, leveraging statistical methods for precise risk assessment.
- Expertise in Logistic Regression and advanced ML models for building application and behavior scorecards to predict default risk for business decisions and capital calculations.
- Experience in calibrating and refining PD/LGD/EAD models to ensure risk adjusted pricing and provisioning.
- Developed forward-looking credit models incorporating macroeconomic factors and point-in-time data to estimate lifetime ECL.
- Gained a strong understanding of RBI's Draft ECL Framework (2025), including staging, SICR assessment, and forward-looking provisions
- Explored key regulatory expectations around PD, LGD, EAD modelling and governance standards aligned with IFRS9 principles.
- Expertise in designing predictive models optimizing customer acquisition strategies and business expansion through advanced data analysis, machine learning, and statistical modeling techniques.
- Demonstrated leadership in managing analytics teams, optimizing task allocation, improving project efficiency and ensuring successful and timely execution.

Functional Area:

- Regulatory Credit Risk Model (PD, EAD, LGD) Development
- Business Model (Application Scorecard/PD Scorecard) Development and Validation
- IFRS9, BASEL IRB

Skills:

- Predictive Analytics, Statistical Modelling (Regression, Classification)
- Machine Learning (Decision Tree,Random Forest, XGBoost)
- Monte Carlo Simulation, R, Python, SQL, SAS Viya
- PowerBI, Excel, SAS Visual Analytics.

Work Experience:

Capgemini, Mumbai — 15 May 2025 - Present

Senior Consultant

- **Developed IRB scorecards** for the mortgage portfolio, behaviour scorecard, and Basel capital charge calculation.
- Led the development of an **Application scorecard** for the auto loan portfolio to assess applicants' creditworthiness, reduce defaults, and improve customer acquisition.
- Built a **Model Validation Accelerator** (MVA) tool using Generative AI and AWS Bedrock to streamline and automate model validation processes.

SBI General Insurance, Thane Mumbai **15 May’23 - 13 May’25**

Analytics Manager

- Developed predictive propensity models for customer retention using SAS Viya and Python, leveraging ETL processes and Exploratory Data Analysis (EDA) to extract, preprocess, and analyze data from diverse sources.
- Designed and developed dashboards on the finalized dataset for motor retention and cross-sell universe using SAS Visual Analytics, enabling in-depth analysis of cross-sell responses and supporting strategic decision-making for future campaigns.

eClerx Services Ltd, Airoli Mumbai **July 2021 - May 2023**

Senior Analyst

- Developed an NLP model using Spacy and Python to extract tagged information from text documents in PDF invoices.Also worked on a Cash Flow (CF) Simulation project to predict revenue for the next 18 months from ongoing company projects.
- Collaborated with the Transaction Reporting team to diagnose and resolve data issues across global regions, implemented rules in a Spark-based rule engine to enhance transaction data quality, and created insightful Power BI visualizations to support decision-making.

Quantafic Business Solutions, Mumbai **Oct 2018 - July 2021**

Data Science Analyst

- Calculated the impact on Expected Credit Loss using R based on changes in PD & LGD for YOY Retail & Non-Retail loans to save underwriting costs.
- Built an internal model using R for non-performing mortgage customers to offer settlement deals based on logistic modeling techniques.
- Led the Grameen KYC & Aadhaar Masking project, automating PDF processing for KYC documents, extracting images, adjusting orientation, and storing data in SQL. Implemented Aadhaar masking to ensure RBI compliance.

Education:

- B.Tech in Computer Science & Engineering (2014) from Dr. K.N. Modi Institute of Engineering & Technology.

Achievements:

- Received the highest Customer Satisfaction Index (CSI) of 100% from the customers.
- Received a rising star award for excellent work for the QCAM android app.
- Certificate of appreciation for exemplary contribution during work from home period.

Projects:

- **Developed a Through-The-Cycle (TTC) Loss Given Default (LGD) model using the Chain Ladder approach [Mar-Oct23]**
  - The project begins with data collection and preprocessing, including historical defaulted loan data, recovery cash flows, and time-to-recovery trends. Using the Chain Ladder method, cumulative recoveries are analyzed across different time periods to project future recoveries. This approach assumes that past recovery patterns hold predictive value for future recoveries.
  - The LGD calculation involves:
    - Constructing development triangles to track cumulative recoveries over time.
    - Applying Chain Ladder factors to extrapolate future recoveries and estimate final LGD.
  - The model is validated through backtesting and stability analysis, ensuring accuracy in predicting long-term recovery rates.

- **Converting TTC LGD to PIT LGD Using Jacob Frye Method [Jul-Aug 23]**
  - Developed a model to convert **Through-The-Cycle (TTC) LGD** to **Point-In-Time (PIT) LGD** as part of credit risk modeling training. The project aimed to adjust LGD estimates to reflect **current economic conditions** rather than long-term averages.
  - Applied the **Jacob Frye method**, incorporating macroeconomic factors such as **GDP growth, unemployment rates, and market trends** to dynamically adjust TTC LGD estimates. This approach improved **real-time risk assessment**.

- **Amortization Schedule Development for Term Loans.[Mar-Aug23]**
  - Developed an **amortization schedule model** to analyze loan repayment structures. The model calculates **monthly principal and interest payments** based on loan amount, interest rate, and tenure, providing a clear breakdown of the **outstanding balance over time**.
  - Also, Developed a model to incorporate **prepayment behavior** into the **amortization schedule** for funded facilities. to adjust loan repayment schedules by factoring in **early repayments** that impact loan balances and interest calculations.
  - Utilized **Python and Excel** to automate schedule generation, enabling insights into **total interest paid, prepayment impact, and loan affordability assessment**. This project enhances risk management by helping lenders evaluate borrower repayment capacity and optimize loan structuring.

- **CCF Computation for Non-Funded Facilities & Computing PIT EAD. [Nov -Dec23]**
  - The project focused on assessing the **potential exposure** of off-balance-sheet items, such as undrawn credit lines, under current economic conditions.
  - Implemented **macroeconomic adjustments** to compute **PIT EAD**, reflecting real-time risk scenarios and the likelihood of utilization. This model enhanced **credit risk forecasting** by accurately capturing exposure under different economic environments

I hereby declare that all the above information is true to the best of my knowledge.
Place: Mumbai Anchit Sharma